

Rchain: A Universal and Efficient Redactable Blockchain Scheme

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Abstract—For the purpose of eliminating harmful data, protecting privacy, updating erroneous data, etc., it is necessary to delete and update the data on blockchains in a safe and controlled way. This paper proposes a redact blockchain scheme based on voting committee. Our scheme achieves controllable redaction because a redaction request is able to be executed only when the major of committee members agree to the request. We use a distributed key generation scheme and a threshold signature scheme to significantly decrease the redaction proof size, which results in small storage overhead. Our scheme is compatible with various consensus mechanisms, supports fast redaction, and has a good scalability. We also formally prove that our scheme satisfies the chain growth, chain quality and redact common prefix security properties. Finally, we develop a proof-of-concept implementation for our scheme, and the experiment results show that the scheme has low time and storage overhead.

Index Terms—blockchains, redactable blockchain, security, efficiency, distributed systems.

I. INTRODUCTION

With the characteristics of decentralization, immutable, and traceability, the blockchain technologies have broad prospects in the fields of industry. The immutable characteristic means that once data is included in a blockchain and becomes stable, it exists forever. On the one hand, the immutability characteristic guarantees the reliability of data of blockchain and is considered to be "source of trust of blockchain." For instance, blockchain technology offers a fully transparent and traceable solution for supply chain management. From raw material procurement to production processing, warehousing and logistics, as well as to final sales, all data is shared in real time on the blockchain, effectively reducing information asymmetry and fraud risks. But on the other hand, the characteristic can also be easily exploited by malicious users to spread incorrect contents with the help of blockchain. Once these incorrect contents are included in a blockchain, they will

exist permanently, and incur large-scale and long-term negative impacts. For example, in several popular blockchain projects, such as Bitcoin [1] and Ethereum [2], computer viruses and botnet codes have been found [3] [4]. There is an urgent need for secure, controllable, and effective methods to eliminate incorrect data contents of blockchain.

From the perspective of protecting user privacy, redactable blockchain technologies help users to delete their sensitive information on blockchain. In May 2018, the new European Union issued General Data Protection Regulation (GDPR) [5] and proposed that users should enjoy the "right to be forgotten". Specifically, under certain conditions, the data owner shall have the right to request the controller to hide or delete its personal data. This right can not be achieved in unmodifiable blockchains, but can realized by adding a modification function in blockchains.

Redactable blockchain technologies can also help to improve executive performance of blockchain by decreasing the ledger length. With the growth of blockchain, the overheads of participants to storage and validate blockchain also increase. By Sep 2023, each full node of Bitcoin or Ethereum needs to take about 500 GB or 1 TB memory to store the blockchain ledger [6] [7], which poses a huge challenge to some users with limited storage and computing power, e.g., mobile phones. Using redactable blockchain technologies, some non-critical historical data can be deleted, thereby reducing the computing and storage costs of users, and improving executive performance of blockchain.

Following above discussions, there exists a strong need to develop redactable blockchain technologies, which will help the healthy development of blockchain, and further promote the wider application of blockchain.

A. Related work

To deal with the problem of harmful data insertions in the blockchain, Ateniese et al. [8] firstly proposed the concept of redactable blockchains and presented a solution in the

This work is supported in part by "A New Technology Architecture for Web3.0 Based on Blockchain" (No. 2023YFB2704200), and in part by Guangxi Science and Technology Major Program (No. GuiKeAA24263017).
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TABLE I
COMPARISON OF OUR SCHEME WITH EXISTING WORKS.

Schemes	Scalability ¹	Public Verifiability	Adversary tolerate	Redaction time/rounds ²	Redaction proof size/KB ³
Ateniese et al	×	×	1/2	/	/
Puddu et al	×	×	1/2	/	/
Deuber et al	✓	✓	1/2 (PoS) or 1/3 (PoW)	801	2420
Thyagarajan et al	✓	✓	1/2 (PoS) or 1/3 (PoW)	801	2420
Li et al	✓	✓	1/2	1	218
Rchain	✓	✓	1/2	1	8

¹ Whether the scheme is suitable for the permissionless blockchain.

² We evaluate the time one redaction complete in synchronous network, where the ratio of honest nodes in the blockchain system $h = 0.6$, and / means not applicable. The voting period is instantiated to 2000 rounds in [11] and [12], and the committee size of our scheme is instantiated to 2000.

³ We evaluate the redaction proof size of 20 redactions, where $h = 0.6$.

permissioned setting. The scheme uses a chameleon hash function [9] to compute the hash pointer of each block. When redacting a block data, a collision for the chameleon hash function can be computed by a trusted party with access to the chameleon trapdoor key, such that the block data can be redacted while maintaining the link relationship between adjacent blocks. However, due to the use of a multi-party secure computation (MPC) protocol, this scheme is more suitable for permissioned blockchain with fixed and small number of participants.

Puddu et al. [10] presented a redactable blockchain based on variable transactions. Each transaction in the blockchain has some different encrypted versions. For each redaction request that meets redaction policy, the miners jointly compute the corresponding decryption key by a MPC protocol, and decrypt the appropriate version of the transaction. However, their scheme does not delete the old transaction history, thus it can not eliminate malicious data in the blockchain. Similarly, the scheme is only suitable for permissioned blockchains due to the need of MPC protocol.

Deuber et al. [11] proposed the first redactable blockchain scheme in the permissionless setting. The scheme relies on a consensus-based voting period, and each redaction request can only be implemented successfully when it receives enough valid votes in the voting period. But the voting period is time consuming, in their application example of Bitcoin, the voting period is set to 1024 rounds, which takes about 7 days to implement a redaction operation. Moreover, by using the chain quality property, the scheme guarantees the majority of voters are honest with negligible failure probability in the voting period. However, when applying the scheme to proof of work (PoW) blockchains, it can only tolerate $< 1/3$ adversary due to the selfish mining attack.

Similarly, Thyagarajan et al. [12] proposed a general redactable blockchain protocol in the permissionless setting, called Reparo. Reparo records redaction history by introducing external data structures and maintains the link relationship between adjacent blocks after performing redactions. However, Reparo also suffers from the problem that the voting period is too long.

Recently, Li et al. [13] proposed a instantly redactable blockchain protocol, where a redaction request can be im-

plemented within a round in the best case. In addition, the voting period of the protocol is escaped from the underlying consensus layer, thus the $1/3$ adversarial ratio is avoided when implementing to PoW blockchains. However, the scheme uses a aggregate signature scheme to aggregate all valid votes to generate the redaction proof, which is included in the chain to prove the validity of each redaction operation. The redaction proof size is linear with the size of the voting committee, as the number of redaction operations increases, the storage overhead could not be neglected.

B. Our contributions

In this paper, we present a redactable blockchain scheme called Rchain in the permissionless setting. The comparison among Rchain and the related schemes are shown in Table I. More specifically, our contribution are summarized as follows.

Controllable redactions and public verifiability. In our scheme, a set of users are randomly selected as the voting committee in each epoch, and the committee selection method guarantees that the major of committee members are honest. Each redaction operation can only be implemented after obtaining enough valid votes from the voting committee in the current epoch (i.e., at least one valid vote from the honest committee members), preventing malicious committee members arbitrarily redacting block data. The votes of each successfully executed redaction operation will be included in the chain, such that each nodes, especially new nodes, can verify the validity of the redaction operation.

General construction and instant redactions. The selection of voting committee is independent to consensus mechanisms, thus our construction can be applied to different blockchains. Moreover, our scheme supports instant redactions. For valid redaction requests, in the synchronous network setting, only one round is needed to collect enough valid votes, which is helpful to correct incorrect block data in time.

Small storage overhead. Our redactable blockchain scheme uses a threshold signature scheme (TSS) to aggregate enough valid votes from the voting committee into a final signature, then a redaction proof that proves the validity of redaction operations is generated and included in the chain, thus all users are able to validate each redaction operation. Due to the use of TSS, the redaction proof size of our scheme is independent

to the committee size, thus it significantly reduces the storage overhead.

Comprehensive performance evaluation. Finally, we develop a proof-of-concept (PoC) implementation for our scheme, and the experiment results show that the scheme is efficient and practical.

II. OVERVIEW OF RCHAIN

This section gives a high level description of our redactable blockchain protocol Rchain, which is depicted in Figure 1.

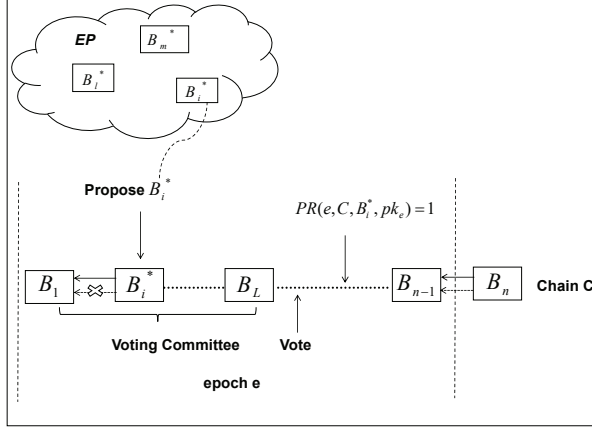


Fig. 1. Overview of Rchain. Each Block is shown as a rectangle and epochs are separated by dashed lines. Each block is linked to its predecessor by two links: the old link (denoted by solid arrow) and the new link (denoted by dashed arrow). $RP(e, C, B_i^*, PK_e) = 1$ denotes that B_i^* obtains enough valid votes from the voting committee of epoch e .

In order to maintain the link relationship between adjacent blocks in the blockchain after performing redaction operations, referring to [11], our redactable blockchain protocol adjusts the structure of each block to be of the form $B_i = \langle s_i, r_i, d_i, y_i, \sigma_i \rangle$, where $s_i = H(s_{i-1}, r_{i-1}, G(d_{i-1}), y_{i-1}, \sigma_{i-1})$ is the hash value of the previous block header, d_i is the block data of B_i , r_i is a random number, $y_i = G(d_0)$ represents the original state of B_i , d_0 is the original block data of B_i without any redaction, H and G are collision-resistant hash functions, σ_i is a signature on message $m = \langle s_i, r_i, d_i, y_i \rangle$. After B_{i-1} is modified, the new link $s_i = H(s_{i-1}, r_{i-1}, G(d_{i-1}), y_{i-1}, \sigma_{i-1})$ is broken, but the old link $s_i = H(s_{i-1}, r_{i-1}, y_{i-1}, \sigma_{i-1})$ still holds, then the link relationship between B_{i-1} and B_i is still regarded to be maintained.

Our protocol is suitable for permissionless blockchain, namely users can freely join or exit the system. Each user owns a pair of long-term public and private keys, and controls one or more nodes simultaneously. It is assumed that the ratio of honest nodes $h = 1/2 + \epsilon$, where $\epsilon > 0$. Our protocol works in K -semi-adaptive corruption model, where the adversary can corrupt a node only after K rounds.

Our redactable blockchain protocol runs in rounds and executes the following steps in each round:

(1) **Propose a redaction.** Each user is able to propose to redact the blockchain. The user who wants to redact a block B_i in the blockchain, generates a candidate redaction block B_i^* that included the new data, and broadcasts B_i^* into the network. In particular, if the user want to delete the data of B_i , it sets the data of B_i^* as null. After receiving B_i^* from the network, each node validates the validity of B_i^* , and adds B_i^* to its editing pool EP if B_i^* is a valid candidate redaction block.

(2) **Generate and broadcast votes.** In each epoch, a set of nodes, namely the leaders of the first L block are selected as the voting committee of this epoch, where the committee size is denoted as S . The committee selection methods guarantees that the ratio of honest committee members is more than $1/2$ (see lemma 1 of section V for more details). Once the committee members are determined, they jointly invoke a DKG protocol to generate their secret and public key shares. Afterwards each committee member generates votes for each candidate redaction block in its editing pools by its secret key, and broadcasts its votes.

(3) **Extend the chain.** At each round j , the current leader creates a new block B_j to extend the chain, and broadcasts the new chain. If the round leader receives more than $1/2S + 1$ valid votes for B_i^* from the current voting committee before generating the new block, it aggregates all these votes into a final signature by a threshold signature scheme, adds the final signature into the data of its new block B_j .

(4) **Redact the chain.** Once the block contain the valid threshold signature on B_i^* is stable in round l , namely the block is buried by k consequent blocks, the current leader replaces B_i in chain with B_i^* , then broadcasts the new chain.

In the above steps, the reason why the final threshold signature needs to be included in the chain is to make all users can public verify the validity of the redaction operation. Notice that the number of malicious committee members is less than $1/2S$, and the voting threshold value of $1/2S + 1$ for executing redaction operations is to prevent malicious committee members from redacting the chain arbitrarily.

III. SPECIFIC CONSTRUCTION OF RCHAIN

In this section, we detail our redactable blockchain protocol Rchain, which is also depicted in Figure 2. Before describing Rchain, we first introduce the voting committee selection method and redaction policy.

Voting committee selection: In each epoch e , the leaders of the first L blocks of epoch e are elected as the voting committee of epoch e , where $L = (\lceil 16 \log(\frac{1}{\delta}) / \epsilon^2 \rceil + 1)$ and δ is a negligible failure probability. The leaders of each epoch are selected by the underlying consensus protocol. The committee selection method guarantees that the ratio of honest members in each committee $\geq 1/2$ (see lemma1 of section V for more details).

Once the committee members of epoch e are determined, all committee members run a $(1/2L + 1, L)$ -DKG protocol to generate the public and secret key shares (pk_i, sk_i) . Afterwards the main public key of TSS needs to be published. Specifically,

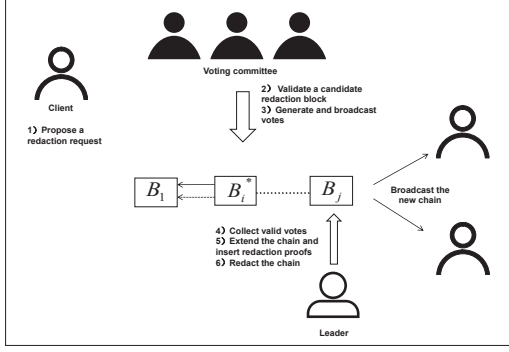


Fig. 2. The overall process of Rchain.

each member generate a signature on the main public key with their long term secret keys and broadcast these signatures. The public key which obtains more than $1/2L$ valid signatures from the committee is accepted by users, which guarantees only the correct main public key is accepted by honest users. Each user uses a local array $PK[e]$ to record the accepted main public key of epoch e .

Redaction policy: We introduce the notion of a redaction policy RP that determines whether a redaction operation to the chain C is valid or not. Specifically, if a block B_j in C is generated during epoch e and has become stable, and B_j contains a valid TSS signature on $H(B_i^*)$ that can be verified successfully with PK_e , then $RP(C, B_i^*, PK_e) = 1$ and we consider that the redacted block B_i^* satisfies the redaction policy, otherwise $RP(C, B_i^*, PK_e) = 0$, where PK_e is the accepted main public key of TSS in epoch e .

Then we describe the main procedures of our redactable blockchain protocol in detail, which is also shown in Figure 2. In each round t of each epoch e , the protocol executes as follows:

1) Propose a redaction request. Any user in the blockchain system can propose to redact a block B_i in the blockchain C , and creates a candidate redaction block B_i^* by algorithm 1. The algorithm replaces the original date d_i of B_i with the new data d_i^* , and then broadcast the candidate redaction block B_i^* into the network. If the user just want to delete all date from B_i , then it sets $d_i^* = \varepsilon$.

Algorithm 1 ProposeRedact (C, i, d_i^*)

The algorithm is run by a user who wants to redact the block at height i in C .

- 1: $B_i = \langle s_i, r_i, d_i, y_i, \sigma_i \rangle$;
- 2: $B_i^* = \langle s_i, r_i, d_i^*, y_i, \sigma_i \rangle$;
- 3: **return** B_i^*

2) Validate a candidate redaction block. When a user receives a candidate redaction block B_i^* from the network, he validates B_i^* by algorithm 2. The algorithm first validates whether B_i^* is a valid block, and then whether the link relationship from B_i^* to B_{i-1} holds and the link relationship

from B_{i+1} to B_i^* holds. If the above conditions are verified successful, then B_i^* is regarded as a valid candidate redaction block and the user adds B_i^* to its own editing pool EP . Otherwise the user just discard B_i^* . Notice that each candidate redaction block has a period of validity. At the beginning of each epoch, each user removes all the candidate redaction blocks from EP .

Algorithm 2 ValidateCand (C, B_i^*)

The algorithm is run by a user who wants to validate a candidate redaction block B_i^* .

- 1: **if** ValidateBlock(B_i^*)=0, **return** 0;
- 2: $B_{i-1} = \langle s_{i-1}, r_{i-1}, d_{i-1}, y_{i-1}, \sigma_{i-1} \rangle$;
- 3: $B_{i+1} = \langle s_{i+1}, r_{i+1}, d_{i+1}, y_{i+1}, \sigma_{i+1} \rangle$;
- 4: **if** $s_i = H(s_{i-1}, r_{i-1}, G(d_{i-1}), y_{i-1}, \sigma_{i-1}) \wedge s_{i+1} = H(s_i, r_i, y_i, y_i, \sigma_i)$, **return** 1;
- 5: **else return** 0

Each users can validate a block B in the chain C by algorithm 3. The algorithm first validates the data included in the block according to the pre-defined rules. It then validate whether the block is produced by a legal leader. If the above conditions are verified successful, the block is regarded as a valid block and the algorithm output 1.

Algorithm 3 ValidateBlock $B = (s, r, d, y, \sigma)$

The algorithm is run by a user who wants to validate a block B in C , where the public key of the user is denoted as pk .

- 1: Validate data d , **if** invalid **return** 0;
- 2: Validate the leader, **if** invalid **return** 0;
- 3: **else return** $V(m = (s, r, d, y), pk, \sigma)$

3) Generate and broadcast votes. Each user checks whether he is a legal voting committee member in the current epoch e . Each legal voting committee member votes for each candidate redaction block B_i^* in its EP . Specially, the committee member generates a valid signature on the hash of B_i^* with its secret key sk_i , and broadcasts its votes to the network. To stimulate the voting committee members to contribute their votes, some incentive mechanisms are needed, e.g., users can choose to include some reward amounts and distribution methods in its new block.

4) Collect valid votes. When a user receives a vote v , he first validates the validity of the vote by algorithm 4, then adds each valid vote into its vote pool VP . In order to prevent a large number of invalid redaction requests, each vote has a period of validity. At the beginning of each epoch, the votes generated in previous epochs become invalid, and each user removes all the votes from VP .

5) Extend chain. In each round, each user tries to extend its local chain C by running algorithm 5, and broadcasts the new chain. Specifically, each user first selects the longest valid chain from each received chain C' as the main chain, then verifies whether he is elected as the round leader. If a user P_i is elected as the round leader, he extends the main chain by generating a new block B_j . If there are enough valid votes for

Algorithm 4 ValidateVote $v = (m, pk, \sigma)$

The algorithm is run by a user who wants to validate a vote v , where the current epoch index is denoted as e .

- 1: **if** the voter of v does not belong to the voting committee in epoch e , then return 0;
- 2: **else return** $V(m, pk, \sigma)$

Algorithm 5 ExtendChain (C, C')

The algorithm is run by each user who wants to extend the underlying chain. The set of valid votes on $H(B_i^*)$ is denoted as V and the number of elements of V is denoted as T . If P_i is elected as the current round leader, then $L(P_i) = 1$, otherwise $L(P_i) = 0$.

- 1: **if** $\text{ValidateChain}(C', \text{len}(C')) = 1 \wedge \text{len}(C') > \text{len}(C)$, **then** $C = C'$;
- 2: **If** $L(P_i) = 0$, **return** null;
- 3: **else if** $T \geq 1/2L + 1$, then $\sigma_i = TR(V)$ and add $< H(B_i^*), \sigma_i >$ into the data of B_j .
- 4: **return** $C || B_j$

a candidate redaction block B_i^* in his own VP , P_i aggregates these votes into a final signature σ_i and adds the redaction proof $< H(B_i^*), \sigma_i >$ into the data of B_j .

Given a chain C' , each user can validate C' by algorithm 6. The algorithm first checks the validity of each block B_i in C' , and then checks whether the link from $B_i = < s_i, r_i, d_i, y_i, \sigma_i >$ to $B_{i-1} = < s_{i-1}, r_{i-1}, d_{i-1}, y_{i-1}, \sigma_{i-1} >$ holds. If $s_i \neq H(s_{i-1}, r_{i-1}, G(d_{i-1}), y_{i-1}, \sigma_{i-1})$ and $s_i = H(s_{i-1}, r_{i-1}, y_{i-1}, y_{i-1}, \sigma_i)$, namely B_{i-1} is an edited block, the algorithm additionally needs to check whether B_{i-1} satisfies the redaction policy.

Algorithm 6 ValidateChain $(C', n = \text{len}(C'))$

The algorithm is run by each user who wants to validate the validity of C' .

- 1: **if** $n = 1$, **return** $\text{ValidateBlock}(B_1)$;
- 2: **else** $i = n$;
- 3: **While** $i > 1$ **do**
- 4: $B_i = < s_i, r_i, d_i, y_i, \sigma_i >$;
- 5: **if** $\text{ValidateBlock}(B_i) = 0$, **return** 0;
- 6: **else if** $s_i = H(s_{i-1}, r_{i-1}, G(d_{i-1}), y_{i-1}, \sigma_{i-1})$, **then** $i = i - 1$
- 7: **else if** $s_i = H(s_{i-1}, r_{i-1}, y_{i-1}, y_{i-1}, \sigma_i) \wedge RP(C, B_{i-1}^*, PK_e) = 1$, **then** $i = i - 1$
- 8: **else return** 0;
- 9: **end While**;
- 10: **return** 1

6) Redact chain. In each round, the leader tries to redact the underlying blockchain by algorithm 7. Specifically, for each candidate redaction block B_l^* in its EP , the leader checks whether $RP(C, B_l^*, PK_e) = 1$ holds. If yes, the leader replaces B_l with B_l^* and broadcasts the new chain.

Algorithm 7 RedactChain (C)

The algorithm is run by the leader of each round, where $C[l]$ denotes the block of height l in chain C , and the current epoch index is e .

- 1: **for** each $B_l^* \in EP$;
- 2: **if** $RP(C, B_l^*, PK_e) = 1$, **then** let $C[l] = B_l^*$;
- 3: **broadcast** C

IV. PERFORMANCE EVALUATION

Experimental Setup. In this section, we develop a proof-of-concept (PoC) implementation to evaluate the performance of Rchain in standard C language. In this experiment, the underlying blockchains simulates the basic function of the blockchain protocol adopted by Cardano [14] and Rchain. Our implementation adopts the ECDSA signature scheme [15] for block proposals. For DKG scheme and threshold signature scheme, we adopt a scalable scheme proposed in [16]. We also assume that the network is synchronous and each round is set to 20 seconds.

Time overhead evaluation. First of all, we evaluate the time overhead of DKG protocol used in our scheme. In figure 3, for different h and δ , we plot the end-to-end time of DKG protocol, which is the sum of the sharing and reconstruction phase times. As h and δ increases, the time overhead of DKG protocol decreases, this is because that less voting committee members need to be selected to run the DKG protocol. However, as shown in figure 3, the time overhead remains tens of second under different h and δ , thus it can be concluded that the DKG protocol only incurs very low time overhead to our redactable blockchain protocol.

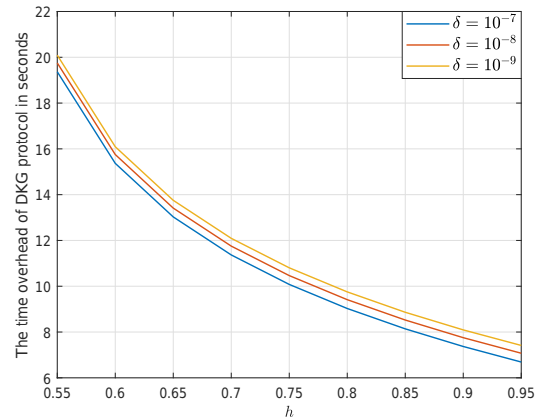


Fig. 3. The time overhead of DKG protocol.

Secondly, for other key steps in Rchain, the time cost is evaluated, where the committee size is fixed as 2000. The experiment conducts several independent experiments for each key step and finally obtains the average time cost. The results are shown in Table II, where the average time for generating a redaction proof represents the average time from collecting

TABLE II
THE TIME COST OF KEY STEPS IN RCHAIN

Average time to generate a vote	~ 7 ms
Average time to validate a vote	~ 1 ms
Average time to generate a redaction proof	~ 584 ms
Average time to validate a redaction proof	~ 94 ms

TABLE III
THE TIME COST OF VOTING PERIOD IN DIFFERENT REDACTABLE BLOCKCHAIN PROTOCOLS

Schemes	$h = 0.9$	$h = 0.8$	$h = 0.6$
the scheme in [11]	201 rounds	401 rounds	801 rounds
the scheme in [13]	1 round	1 round	1 round
Rchain	1 round	1 round	1 round

enough valid votes to generating a redaction proof. As shown in Table II, the average time of these key steps are less than 1 second. Compared to the time interval between two blocks (it takes 20 seconds to produce a new block in Cardano), we can conclude that these key steps only add a small time overhead, thus the issue of new blocks will not be delayed.

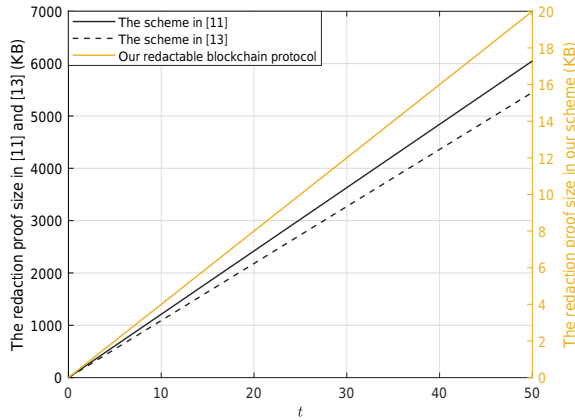


Fig. 4. The redaction proof size in different redactable blockchain protocols.

In the third experiment, for valid redaction requests, we compare the time cost of time-consuming voting period in different redactable blockchain protocols under different h , where the committee size L is fixed as 2000. As shown in table III, under the condition of synchronous network, it only takes one round (about 20 seconds) to collect enough votes in Rchain and the scheme in [13]. However, in [11], the time cost of voting period is linear to h . Thus we can conclude that both Rchain and [13] support instant redaction operations.

Storage overhead evaluation. The last experiment evaluates the storage cost of Rchain. We fix $h = 0.6$ and $L = 2000$, and compare the redaction proof size in different schemes under different t , where t is the number of redaction operations. As shown in figure 4, compared with [11] and [13], Rchain achieves much smaller redaction proof size, which results in lower storage overhead. For example, when $t = 5$,

the redaction proof size in Rchain only accounts for about 0.4% of that in the other two schemes. When t increases, the advantages of Rchain will become more obvious.

V. CONCLUSION

This paper proposes a redactable blockchain scheme that can be used to remove harmful data, protect user privacy, modify erroneous data, and improve blockchain system performance. Based on the voting committee selection and TSS, our scheme has the advantages of supporting different blockchains, achieving controllable and instant redactions, having small storage overhead. To provide strong security, one of our further works could be to design a redactable blockchain protocol in the fully-adaptive corruption model, while preserving instant redaction and small redaction proof size.

VI. ACKNOWLEDGEMENTS

This work is supported in part by “A New Technology Architecture for Web3.0 Based on Blockchain” (No. 2023YF-B2704200), and in part by Guangxi Science and Technology Major Program (No.GuiKeAA24263017).

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